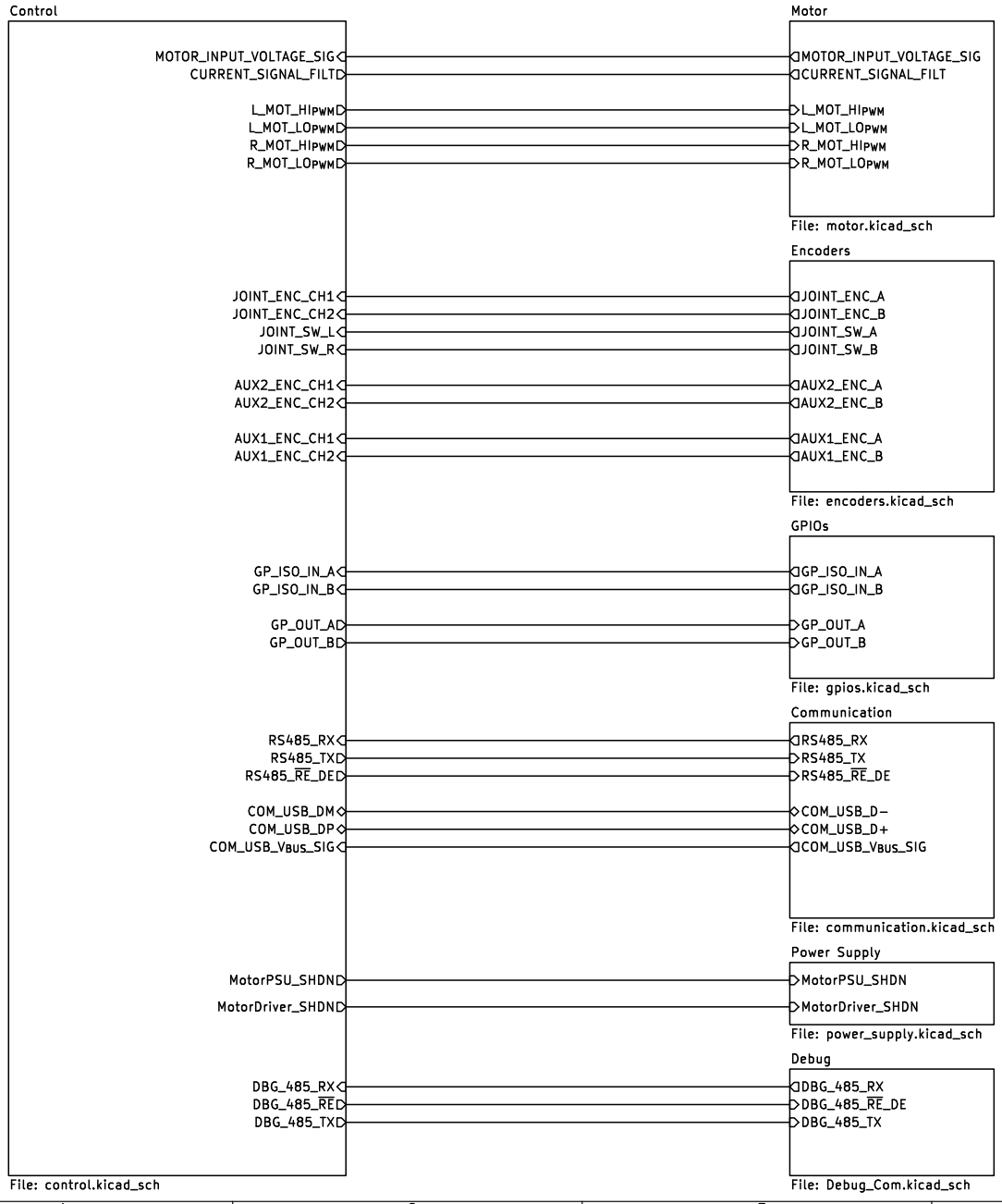
System overview



H1 H4
H3 H2

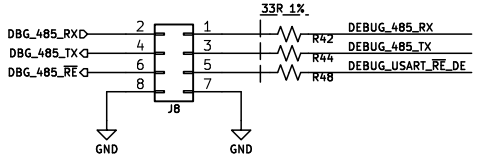
Single Joint Controller Board
Designed by Alves, T. G.

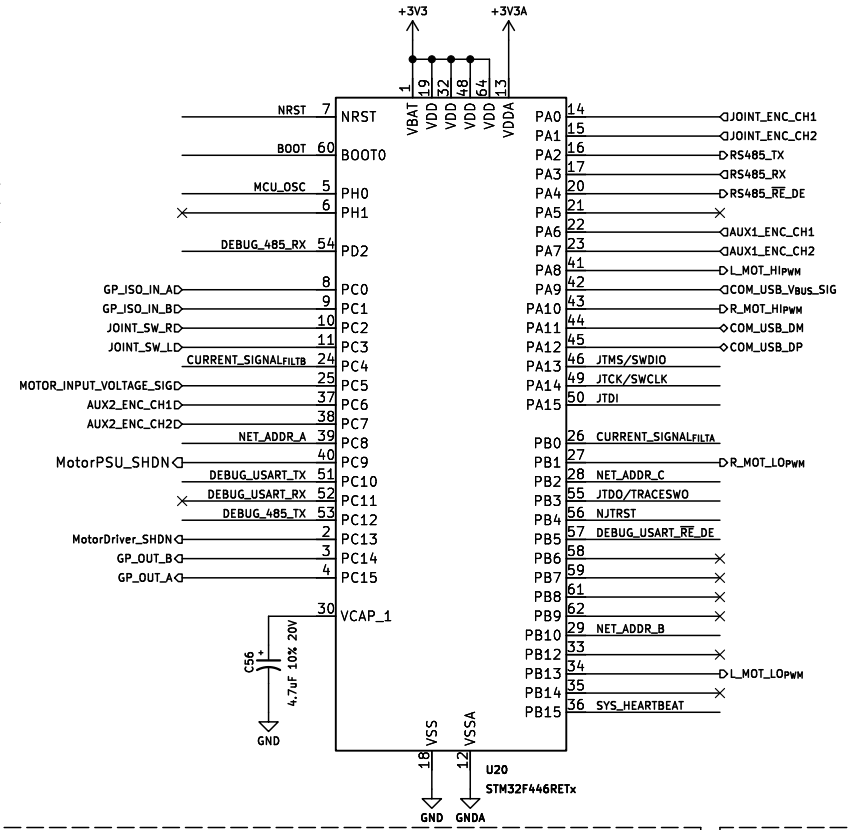
Sheet: /
File: hardware_v2.kicad_sch

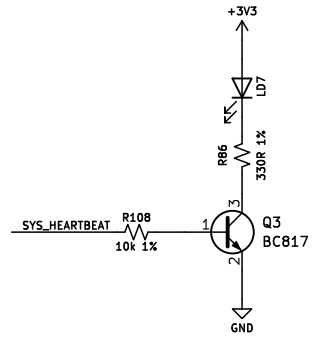
Title: System overview

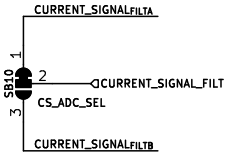
Size: A4	Date: 2024-05-06	**Rev: 2.0**
KiCad E.D.A. 8.0.2		Id: 1/8

Microcontroller





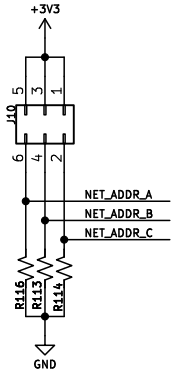




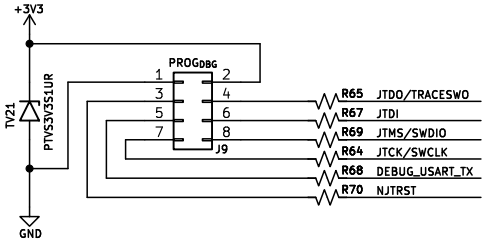
Network Address

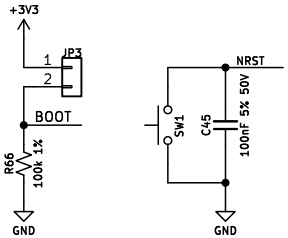
BINARY CODED:

C B A -- VALUE
0 0 0 -- 0
0 0 1 -- 1
0 1 0 -- 2
0 1 1 -- 3
...
1 1 1 -- 7

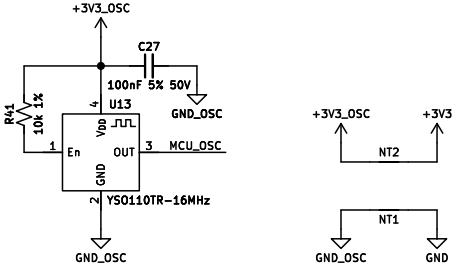


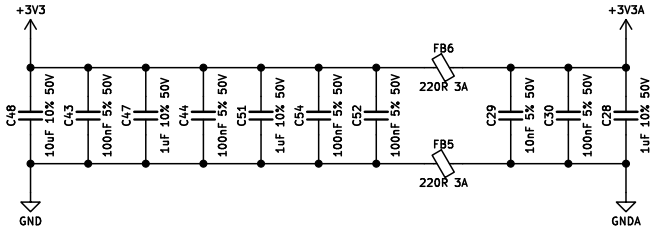
Debug/Prog



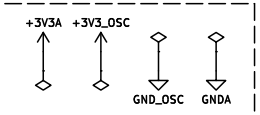


Oscillator





Power flags



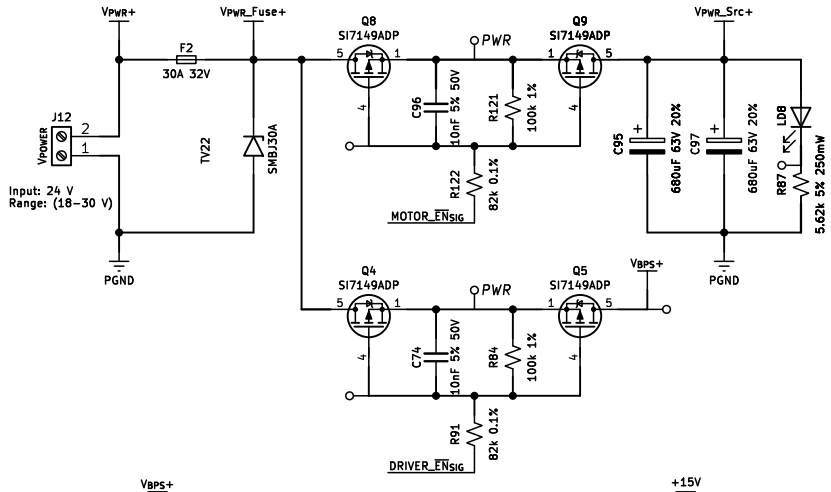
Single Joint Controller Board
Designed by Alves, T. G.

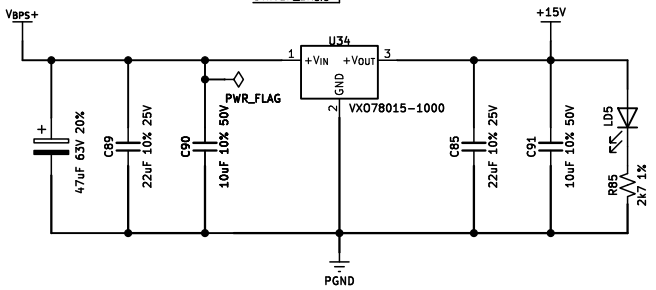
Sheet: /Control/
File: control.kicad_sch

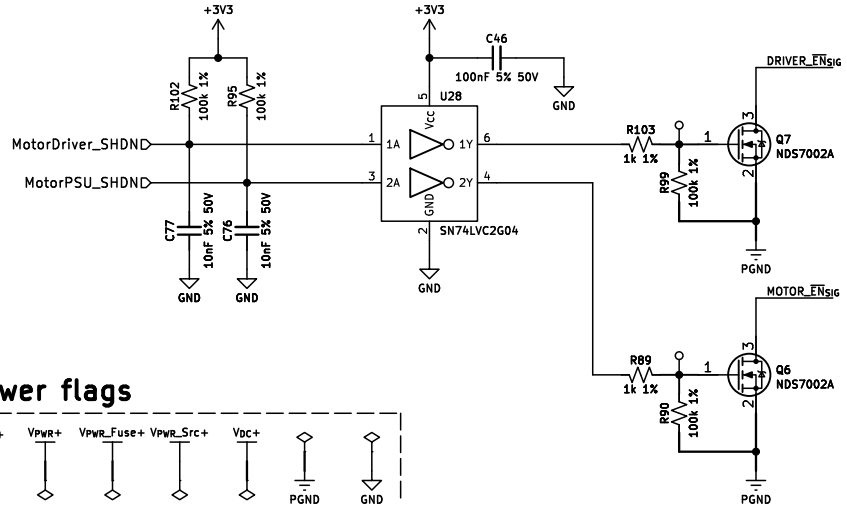
Title: SJC Control

Size: A4	Date: 2024-05-06	**Rev: 2.0**
KiCad E.D.A. 8.0.2		Id: 2/8

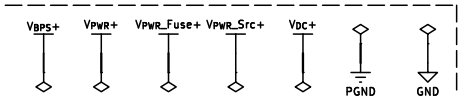
Power Inputs



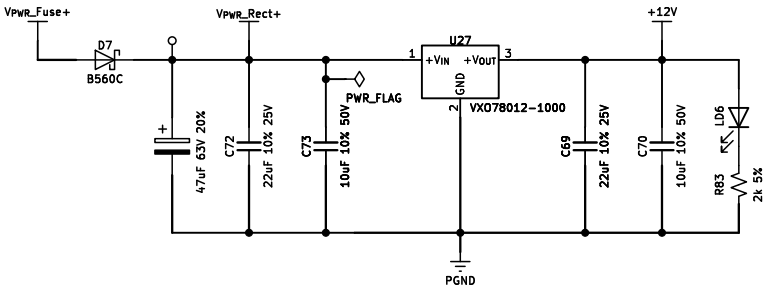


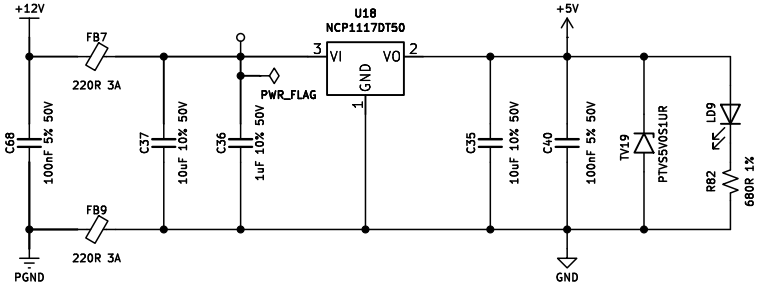


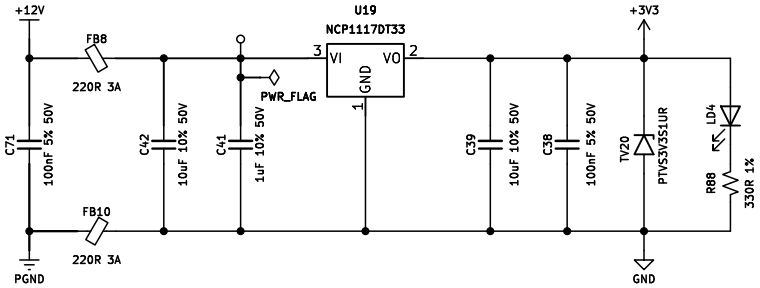
Power flags



Logic Inputs







Single Joint Controller Board
Designed by Alves, T. G.

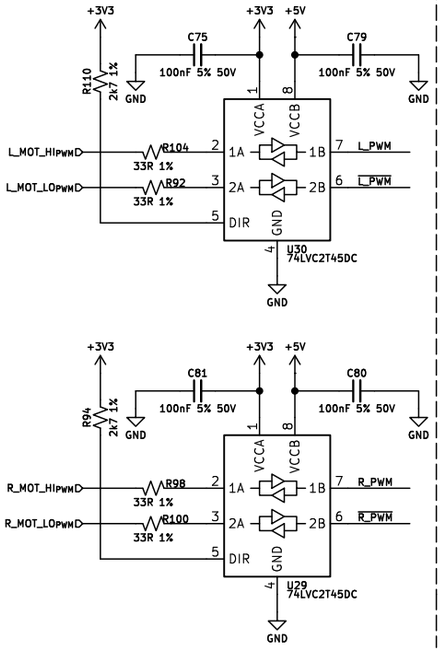
Sheet: /Power Supply/
File: power_supply.kicad_sch

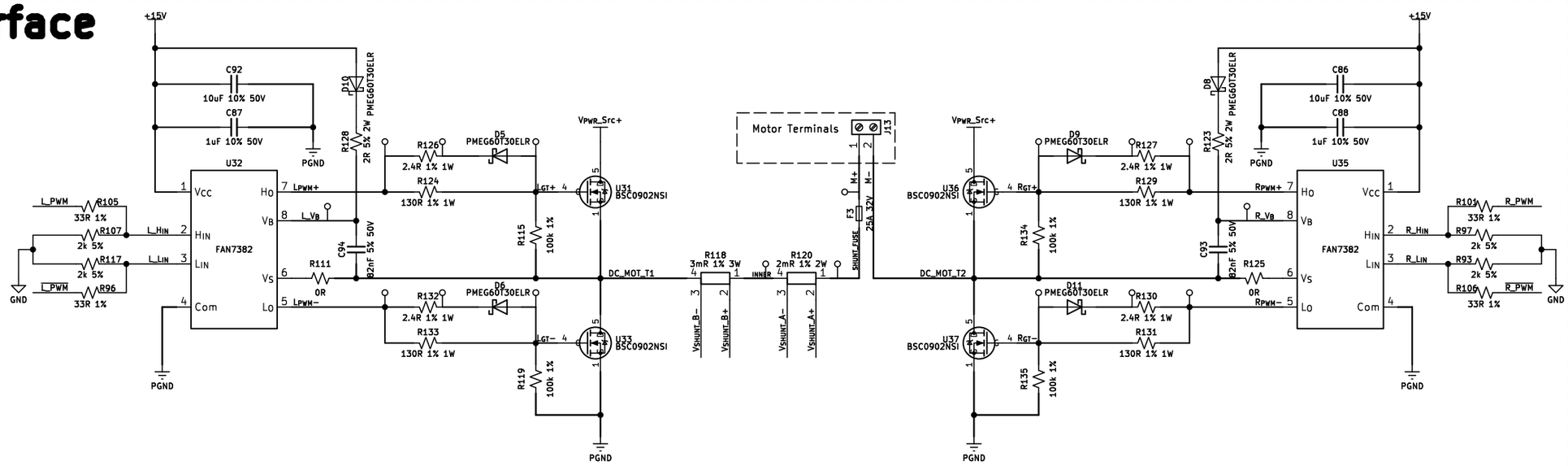
Title: SJC Power Supply

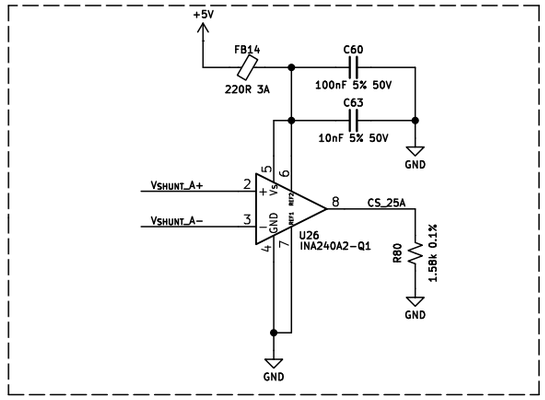
Size: A4	Date: 2024-05-06	**Rev: 2.0**
KiCad E.D.A. 8.0.2		Id: 3/8

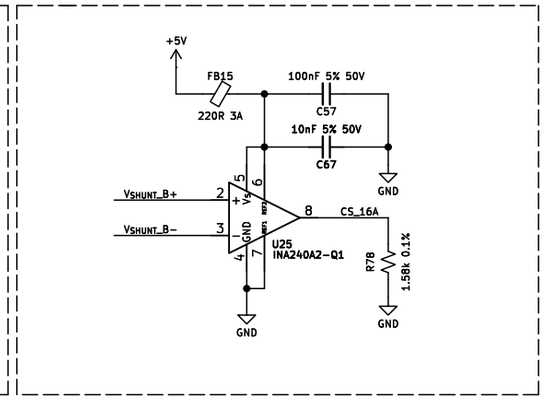
Servomotor interface

PWM Signals

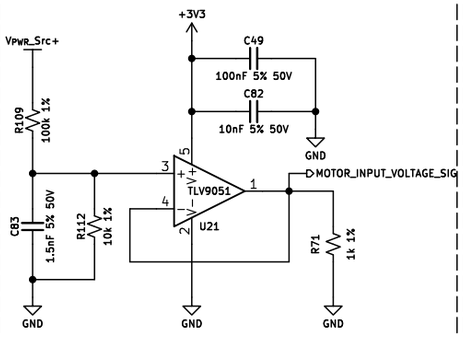




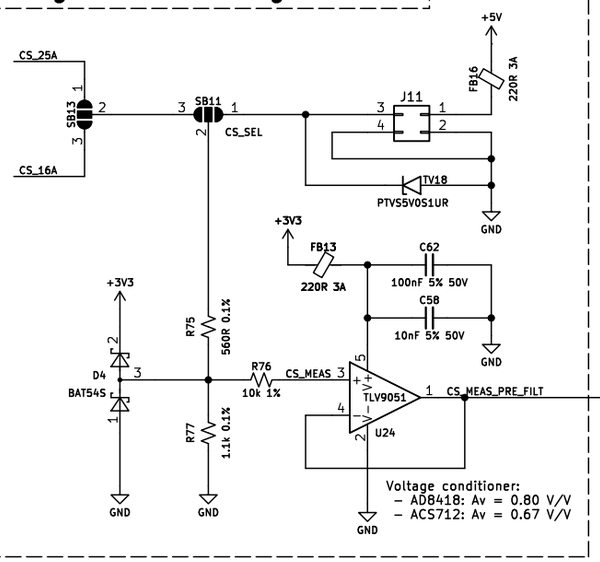


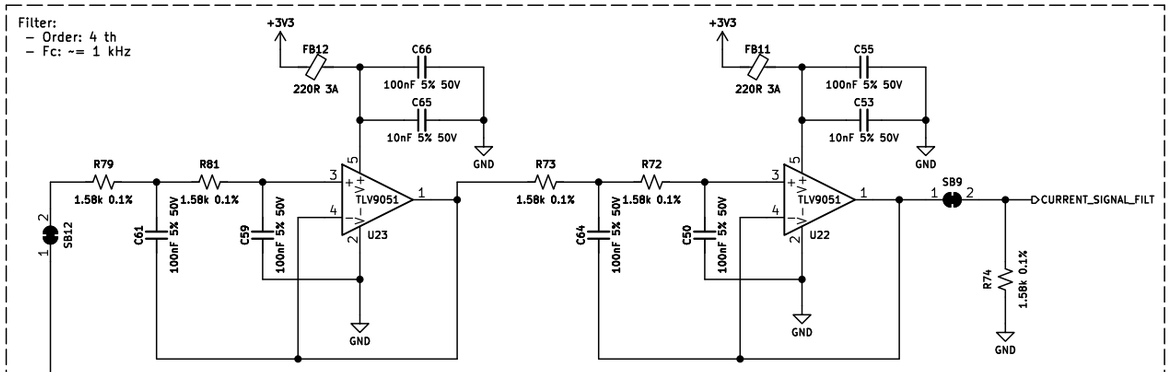


H−Bridge input voltage sense



Current signal conditioning interface





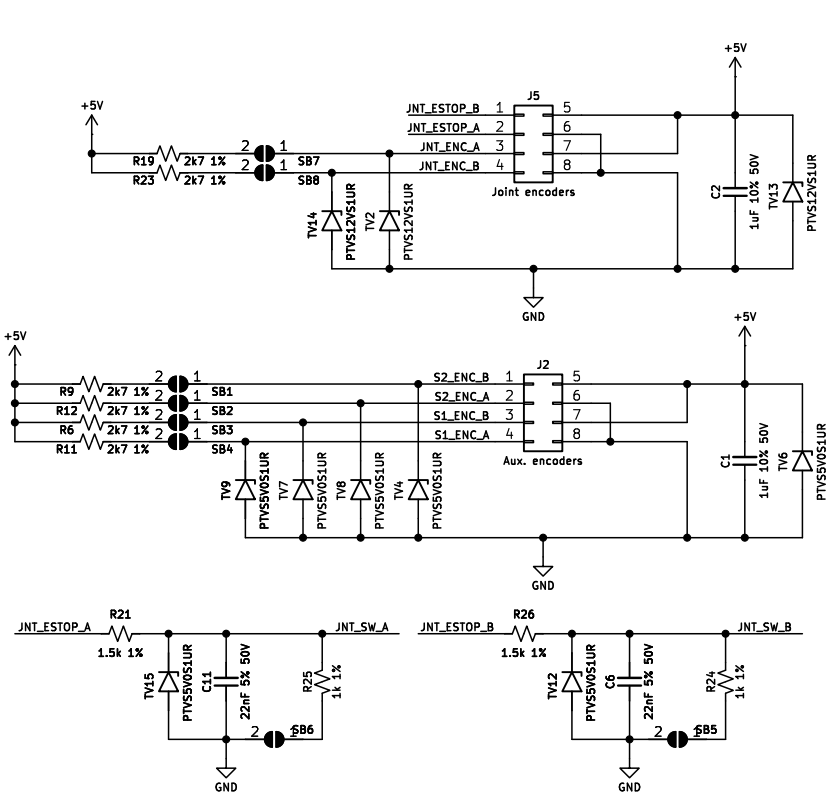
Single Joint Controller Board
Designed by Alves, T. G.

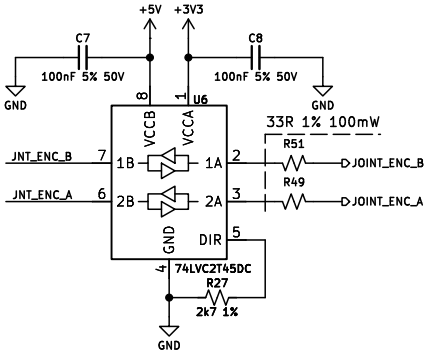
Sheet: /Motor/
File: motor.kicad_sch

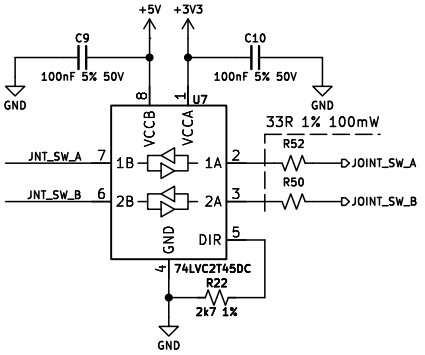
Title: SJC Servomotor

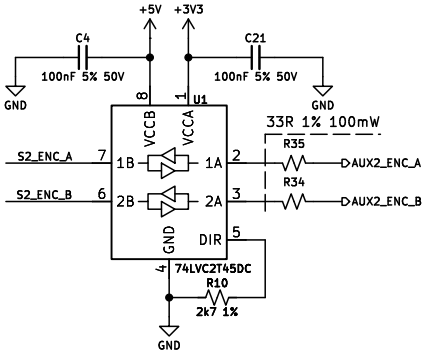
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KiCad E.D.A. 8.0.2		Id: 4/8

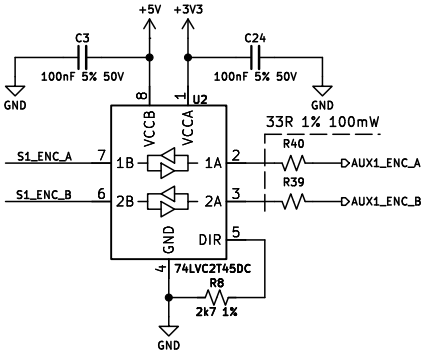
Encoder interfaces











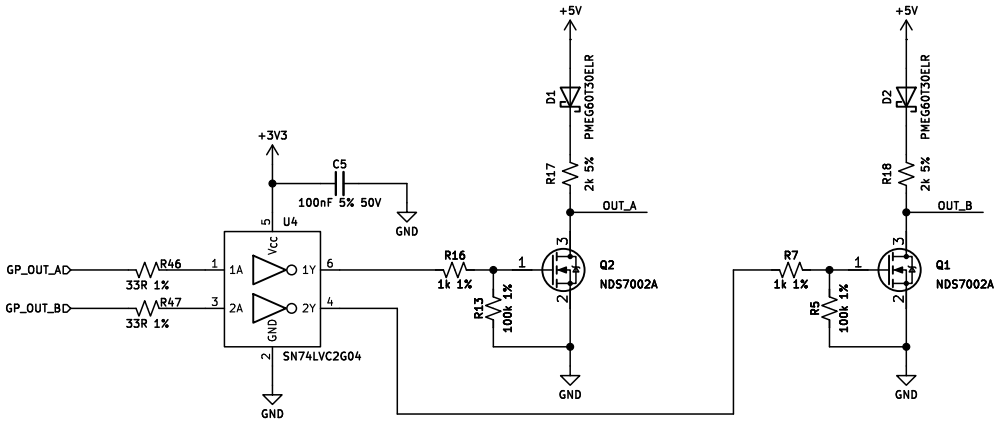
Single Joint Controller Board
Designed by Alves, T. G.

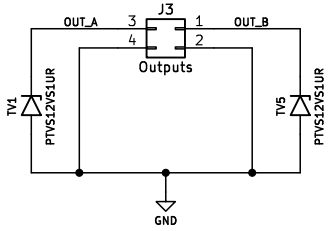
Sheet: /Encoders/
File: encoders.kicad_sch

Title: SJC Encoders

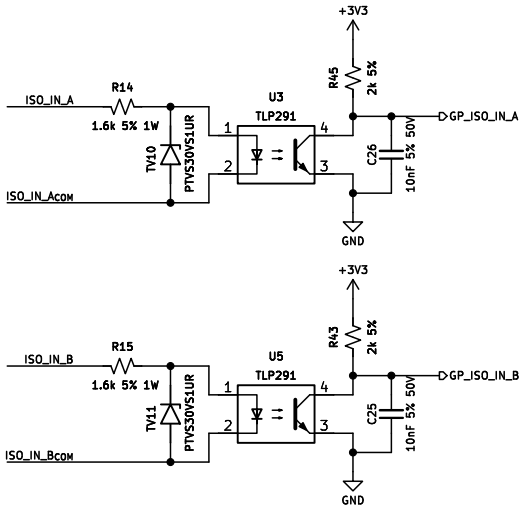
Size: A4	Date: 2024-05-06	**Rev: 2.0**
KiCad E.D.A. 8.0.2		Id: 5/8

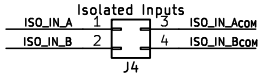
Digital Outputs





Isolated Inputs





Single Joint Controller Board
Designed by Alves, T. G.

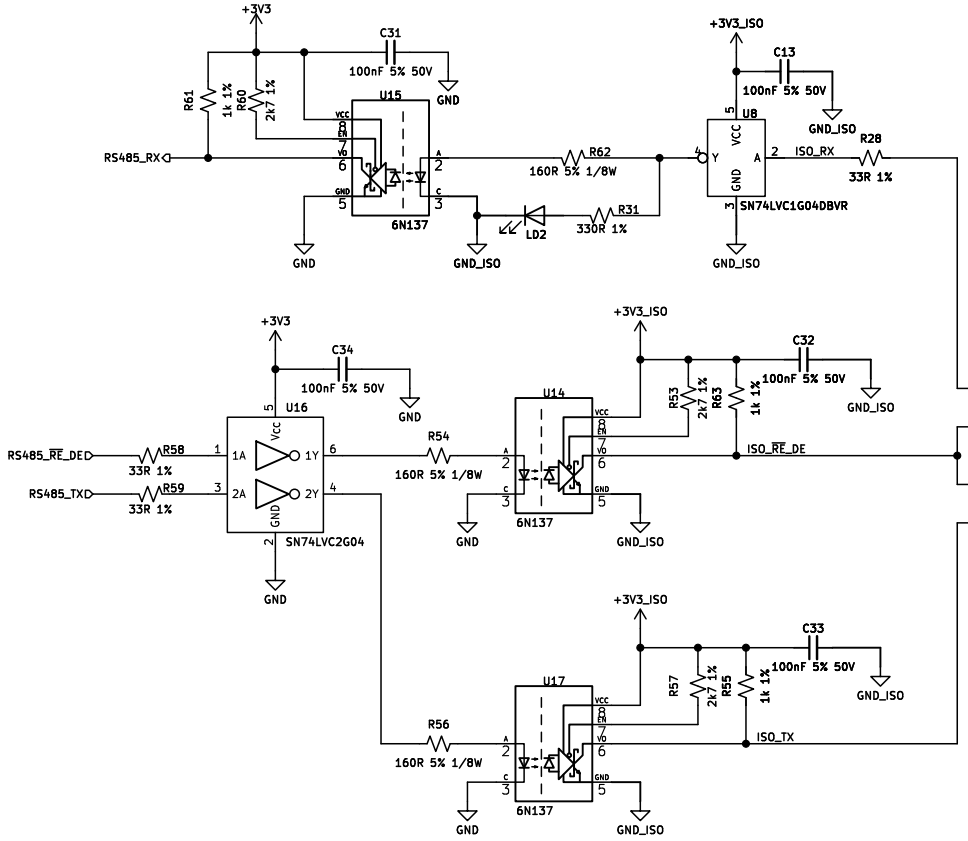
Sheet: /GPIOs/
File: gpios.kicad_sch

Title: SJC I/Os

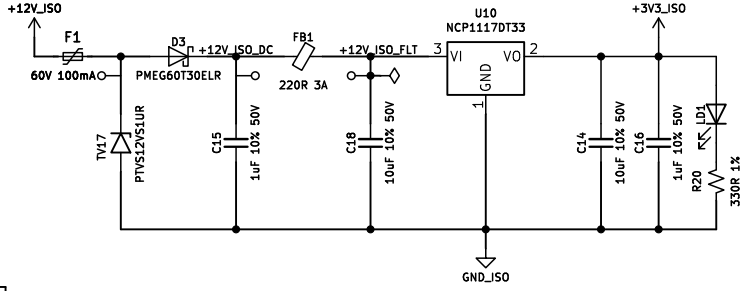
Size: A4	Date: 2024-05-06	**Rev: 2.0**
KiCad E.D.A. 8.0.2		Id: 6/8

Communication

RS-485 (ISO)

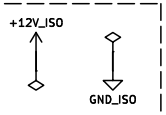


BUS power supply

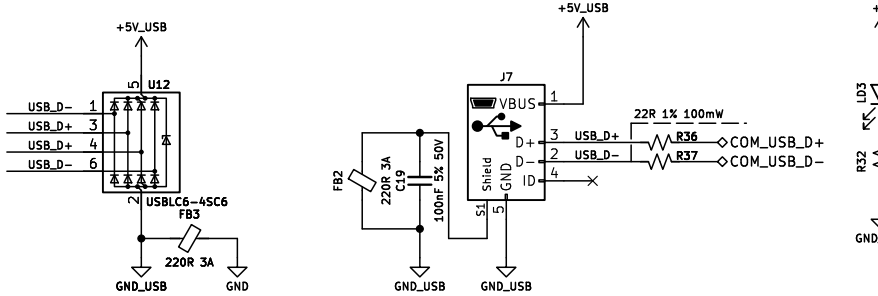


RS-485
- Input: 12 V
- Range: 9-13.3 V

Power flags



USB

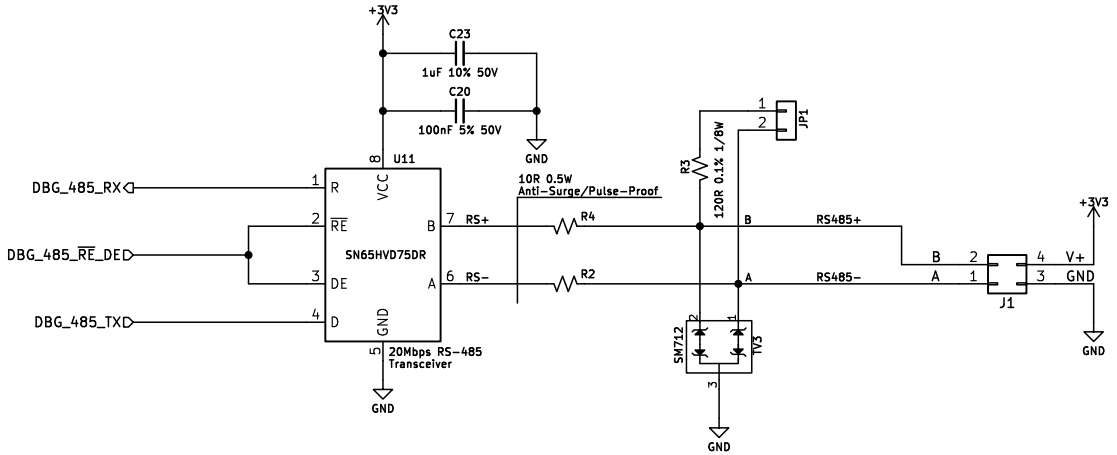


Single Joint Controller Board
Designed by Alves, T. G.

Sheet: /Communication/
File: communication.kicad_sch

Title: SJC Communication

Size: A4	Date: 2024-05-06	Rev: 2.0
KiCad E.D.A. 8.0.2		Id: 7/8



Single Joint Controller Board
Designed by Alves, T. G.

Sheet: /Debug/
File: Debug_Com.kicad_sch

Title: Data Logger Debug Output

Size: A4	Date: 2024-05-06	**Rev: 2.0**
KiCad E.D.A. 8.0.2		Id: 8/8